

EE263 Homework 6
Summer 2026

1. Norm expressions for quadratic forms. Let $f(x) = x^T A x$ (with $A = A^T \in \mathbb{R}^{n \times n}$) be a quadratic form.

- a) Show that f is positive semidefinite (*i.e.*, $A \geq 0$) if and only if it can be expressed as $f(x) = \|Fx\|^2$ for some matrix $F \in \mathbb{R}^{k \times n}$. Explain how to find such an F (when $A \geq 0$). What is the size of the smallest such F (*i.e.*, how small can k be)?
- b) Show that f can be expressed as a difference of squared norms, in the form $f(x) = \|Fx\|^2 - \|Gx\|^2$, for some appropriate matrices F and G . How small can the sizes of F and G be?

2. Properties of symmetric matrices. In this problem P and Q are symmetric matrices. For each statement below, either give a proof or a specific counterexample.

- a) If $P \geq 0$ then $P + Q \geq Q$.
- b) If $P \geq Q$ then $-P \leq -Q$.
- c) If $P > 0$ then $P^{-1} > 0$.
- d) If $P \geq Q > 0$ then $P^{-1} \leq Q^{-1}$.
- e) If $P \geq Q$ then $P^2 \geq Q^2$.

Hint: you might find it useful for part (d) to prove $Z \geq I$ implies $Z^{-1} \leq I$.

3. Squareroot and logarithm of a (diagonalizable) matrix. Suppose that $A \in \mathbb{R}^{n \times n}$ is diagonalizable. Therefore, an invertible matrix $T \in \mathbb{C}^{n \times n}$ and diagonal matrix $\Lambda \in \mathbb{C}^{n \times n}$ exist such that $A = T \Lambda T^{-1}$. Let $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$.

- a) We say $B \in \mathbb{R}^{n \times n}$ is a squareroot of A if $B^2 = A$. Let μ_i satisfy $\mu_i^2 = \lambda_i$. Show that $B = T \text{diag}(\mu_1, \dots, \mu_n) T^{-1}$ is a squareroot of A . A squareroot is sometimes denoted $A^{1/2}$ (but note that there are in general many squareroots of a matrix). When λ_i are real and nonnegative, it is conventional to take $\mu_i = \sqrt{\lambda_i}$ (*i.e.*, the nonnegative squareroot), so in this case $A^{1/2}$ is unambiguous.
- b) We say B is a logarithm of A if $e^B = A$, and we write $B = \log A$. Following the idea of part a, find an expression for a logarithm of A (which you can assume is invertible). Is the logarithm unique? What if we insist on B being real?

4. Ellipsoids.

- a) Write a function that, given a 2×2 real, positive definite symmetric matrix $A > 0$, plots the ellipse

$$E = \{ x \in \mathbb{R}^2 \mid x^T A x = 1 \}$$

Make sure that your plot is shown so that horizontal and vertical lengths are the same, that is, with aspect ratio 1. Turn in your code.

Julia hint: use the following to draw a plot with correct aspect ratio.

Using Plots; `plot(x, y, aspect_ratio=:equal)`

b) Use your code to plot the ellipsoid for the matrix

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

c) Use your code to plot the ellipsoid for the matrix

$$A = \begin{bmatrix} 0.2 & -0.1 \\ -0.1 & 0.4 \end{bmatrix}$$

On your plot, also show semiaxes.

d) Consider an estimation problem, where we have three sensors, define by $b_i \in \mathbb{R}^2$ for $i = 1, 2, 3$. We measure $y_i = b_i^\top x$. The vectors b_i are

$$b_1 = \begin{bmatrix} 0.89 \\ 0.45 \end{bmatrix} \quad b_2 = \begin{bmatrix} 0.45 \\ 0.89 \end{bmatrix} \quad b_3 = \begin{bmatrix} -0.71 \\ 0.71 \end{bmatrix}$$

Plot the set of $x \in \mathbb{R}^2$ for which $\|y\| \leq 1$. On your plot, show also the b_i (that is, plot a line from the origin to b_i).

5. Recovering an ellipsoid from boundary points. You are given a set of vectors $x^{(1)}, \dots, x^{(N)} \in \mathbb{R}^n$ that are thought to lie on or near the surface of an ellipsoid centered at the origin, which we represent as

$$\mathcal{E} = \{x \in \mathbb{R}^n \mid x^\top A x = 1\},$$

where $A = A^\top \in \mathbb{R}^{n \times n} \geq 0$. Your job is to recover, at least approximately, the matrix A , given the observed data $x^{(1)}, \dots, x^{(N)}$. Explain your approach to this problem, and then carry it out on the data given in the mfile `ellip_bdry_data.json`. Be sure to explain how you check that the ellipsoid you find is reasonably consistent with the given data, and also that the matrix A you find does, in fact, correspond to an ellipsoid. To simplify the explanation, you can give it for the case $n = 4$ (which is the dimension of the given data). But it should be clear from your discussion how it works in general.

6. (Optional) Spectral mapping theorem. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is analytic, *i.e.*, given by a power series expansion

$$f(u) = a_0 + a_1 u + a_2 u^2 + \dots$$

(where $a_i = f^{(i)}(0)/(i!)$). (You can assume that we only consider values of u for which this series converges.) For $A \in \mathbb{R}^{n \times n}$, we define $f(A)$ as

$$f(A) = a_0 I + a_1 A + a_2 A^2 + \dots$$

(again, we'll just assume that this converges).

Suppose that $Av = \lambda v$, where $v \neq 0$, and $\lambda \in \mathbb{C}$. Show that $f(A)v = f(\lambda)v$ (ignoring the issue of convergence of series). We conclude that if λ is an eigenvalue of A , then $f(\lambda)$ is an eigenvalue of $f(A)$. This is called the *spectral mapping theorem*.

To illustrate this with an example, generate a random 3×3 matrix, for example using `A=randn(3)`. Find the eigenvalues of $(I + A)(I - A)^{-1}$ by first computing this matrix, then finding its eigenvalues, and also by using the spectral mapping theorem. (You should get very close agreement; any difference is due to numerical round-off errors in the various computations.)